CIF Robustness Checks

Technical Appendix (Methods, Counts, Diagnostics, Outputs)

Data audit and preprocessing

- Duplicates: true duplicate rows removed; ID collisions resolved by enforcing one record per child using a transparent rule (retain the most complete record on key covariates; tie-break by earliest record).
- Variable harmonization: standardized names (e.g., I_to_N_ratio; FaInc) and removed non-analytic identifiers/administrative fields.

Missing data and multiple imputation (covariates only)

- Multiple imputation performed on covariates only (M = 5 datasets). Treatment (FaInc) and outcomes were not imputed.
- Numeric covariates (AgeM, I_to_N_ratio, educat): IterativeImputer (MICE-style) with posterior draws.
- Categorical covariates: program, gender, ethnic; ethnicity collapsed where rare levels (< ~3%) were grouped as Other.
- After consolidation and MI handling decisions, each imputed dataset used for matching contained 1,840 cases (FaInc = 192; controls = 1,648).

```
FaInc==0:
  All three missing        : 0
  Missing on at least one  : 176
  No missing on these three: 1648

Per-variable missing among FaInc==0:
I_to_N_ratio    85
ethnic           7
educat          86
dtype: int64
```

```
FaInc==1:
  All three missing        : 0
  Missing on at least one  : 20
  No missing on these three: 172

Per-variable missing among FaInc==1:
I_to_N_ratio     7
ethnic           0
educat          14
dtype: int64
```

Propensity score model and matching

- Covariates in PS model: program, gender, ethnic_collapsed, AgeM, I_to_N_ratio, educat.
- Encoding: one-hot for categorical covariates; flexible age modeling using a natural cubic spline with 5 knots inside the PS model.
- Propensity model: logistic regression.
- Matching: 1:1 nearest neighbor with replacement on the logit propensity; caliper = 0.2 SD of logit PS.
- Exact matching: gender and ethnic_collapsed (balance is exact by design).

Balance diagnostics

- Matched pairs per imputed set: 190–192 (unmatched treated: 0–2 per set).
- Balance after matching: max |SMD| typically ≤ 0.043; worst-case across sets ≤ 0.070; pre-matching max |SMD| ≈ 0.280.
- Full balance tables and top post-matching imbalances by MI set are provided in the separate Balance Tables document.

Outcome construction and missingness

- PALS harmonization: two versions of PALS-PreK (pals, npals) harmonized by taking the first non-missing value; each child contributes one score.
- Outcome availability (typical/median across MI sets): Social–emotional indicators had near-complete data; academic indicators showed small missingness for alphabet knowledge, sight words, and PALS; math indicators had minimal missingness.

Outcome missingness across matched sets (median counts):

	median_nonmiss	median_miss
PALS	375	5
alphknow	377	3
cooperat	380	0
geometry	379	1
initiat	380	0
measures	379	1
numbers	379	1
selfcont	380	0
sightwrd	378	2
social	380	0

Outcome models: frequentist

- Primary: GEE (Gaussian, identity) with cluster-robust SEs by match_set (matched pair).
- Fallback: OLS with cluster-robust SEs by match_set where mixed-effects estimation would be ill-conditioned; results materially indistinguishable from GEE.
- Covariate adjustment (doubly robust): models included FaInc plus C(program), C(ethnic_collapsed), AgeM, I_to_N_ratio, educat (gender and ethnicity exact-matched).

Representative pooled GEE estimates (FaInc effect; 95% CI)

- Self Control: 0.027 [−0.986, 1.040]
- Cooperation: 0.055 [−0.567, 0.677]
- Social Skills: 0.146 [−0.480, 0.772]
- Initiative: 0.230 [−0.370, 0.829]
- Alphabet Knowledge: −0.417 [−2.276, 1.442]
- Sight Words: −0.136 [−0.543, 0.271]
- PALS PreK: −2.189 [−7.153, 2.774]
- Numbers: −0.849 [−3.342, 1.643]
- Geometry: 0.372 [−0.465, 1.208]
- Measures: 0.185 [−0.192, 0.563]

Bayesian SEM (primary robustness)

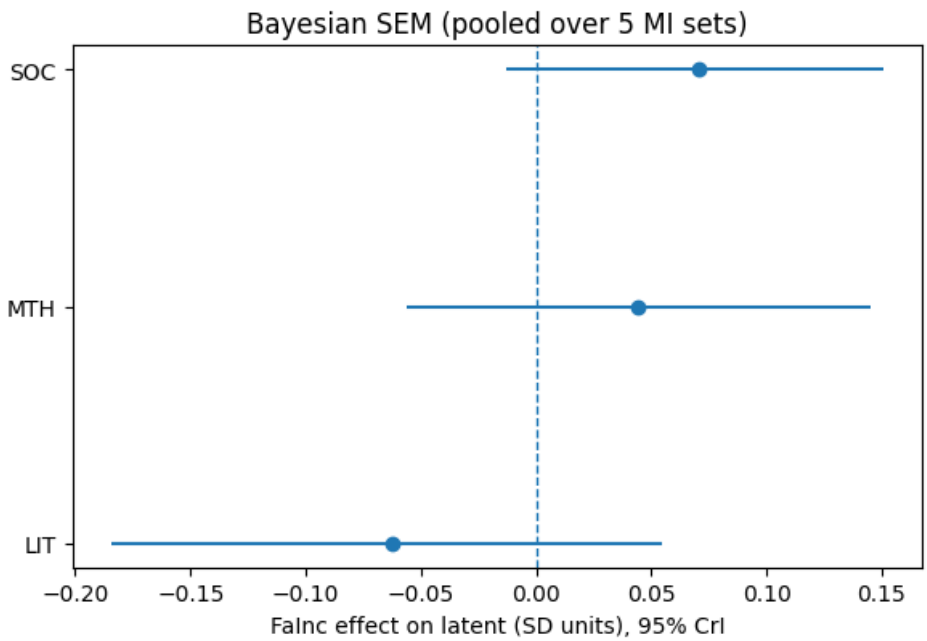
- Measurement model: Social ← selfcont, cooperat, social, initiat; Literacy ← alphknow, PALS, sightwrd; Math ← numbers, geometry, measures.
- Structural model: each latent regressed on FaInc and covariates (program, ethnic_collapsed, AgeM, I_to_N_ratio, educat).
- Clustering: random intercept by match_set for each latent factor.
- Missing outcomes: handled via Bayesian FIML (within-model imputation from the posterior predictive distribution).
- Standardization: indicators z-scored within each MI set for stable priors and interpretable SD-scale effects.

Posterior means for the FaInc effect on Social, Literacy, and Math latent factors were near zero, and the 95% credible intervals spanned zero for all three factors—convergent with the frequentist results.

```
Pooled Bayesian SEM — FaInc effects on latents (z-units):
  latent   term      mean  hdi_2.5%  hdi_97.5%
0    SOC  FaInc  0.070465 -0.012746   0.150759
1    LIT  FaInc -0.062141 -0.184558   0.054660
2    MTH  FaInc  0.044442 -0.056206   0.145382

Diagnostics for FaInc rows (pooled):
  latent  rhat  ess_bulk  ess_tail
0    SOC  1.04     143.0    5801.0
1    LIT  1.03     270.0    4580.0
2    MTH  1.01     669.0    6894.0
```

Figure A. Bayesian SEM robustness results pooled over M = 5 covariate-imputed matched datasets: posterior mean and 95% credible interval for the FI (FaInc) effect on each latent factor (SD units). Vertical line at 0 indicates no effect.



Fitting and pooling

- Model fit separately on each of the 5 MI-matched datasets; diagnostics indicated stable sampling with no reported divergences in the documented runs.
- Pooling approach: concatenate posteriors across MI sets (Bayesian analogue to MI pooling) and summarize FI effects on each latent factor in SD units.

Implementation notebook (Colab)

- Colab notebook (Python implementation) documents the end-to-end pipeline: data audit (duplicate-ID handling), covariate-only multiple imputation (M=5), propensity score estimation with flexible age modeling, 1:1 nearest-neighbor matching with replacement and caliper on logit PS, outcome re-estimation with clustering, and Bayesian latent-variable modeling. The notebook primarily serves as a reproducibility artifact and code appendix.
- Implementation details visible in the notebook include use of common scientific Python tools (e.g., pandas/numpy), scikit-learn components for imputation and feature engineering (e.g., IterativeImputer, one-hot encoding, spline/feature transforms), and Bayesian modeling tools (PyMC/ArviZ) for the Bayesian SEM. (If you want exact package versions, add them from the notebook header/output.)

Outputs (files produced)

- Per-set balance tables (pre/post SMD) and top-imbalance summaries.
- Outcome histograms per MI set.
- outcome_models_results.csv and pooled forest plot for GEE estimates.
- Bayesian SEM pooled effects forest plot (latent SD-scale).